

Plasma carotenoids, retinol and tocopherol levels and the risk of ovarian cancer.

OBJECTIVE: We investigated the relation between plasma carotenoids, retinol and tocopherol levels and ovarian cancer risk in Korean women. **DESIGN:** Hospital-based case-control study. **SETTING:** Six tertiary medical institutes in Korea. **POPULATION:** Forty-five epithelial ovarian cancers and 135 age-matched controls. **METHODS:** Preoperative plasma concentrations of beta-carotene, lycopene, zeaxanthin plus lutein, retinol, alpha-tocopherol, and gamma-tocopherol were measured by reverse-phase, gradient high-pressure liquid chromatography. **MAIN OUTCOME MEASURES:** Odds ratios (OR) and 95% confidence intervals (95%CI) were estimated by tertiles to evaluate the effect of micronutrients on endometrial cancer risk after adjustment for body mass (BMI) index, menopause, parity, oral contraceptive use, smoking status, and alcohol consumption status. **RESULTS:** Women in the highest tertile for beta-carotene had 0.12-times the risk of ovarian cancer of in the lowest tertile (OR 0.12; 95%CI 0.04-0.36). Women with the highest tertiles of lycopene (OR 0.09; 95%CI 0.03-0.32), zeaxanthin/lutein (OR 0.21; 95%CI 0.09-0.52), retinol (OR 0.45; 95%CI 0.21-0.98), alpha-tocopherol (OR 0.23; 95%CI 0.10-0.53) and gamma-tocopherol (OR 0.28; 95%CI 0.11-0.70) had lower risk of ovarian cancer than women in the lowest tertiles. Results were consistent across strata of socio-epidemiologic factors. **CONCLUSIONS:** Micronutrients, specifically ss-carotene, lycopene, zeaxanthin, lutein, retinol, alpha-tocopherol, and gamma-tocopherol, may play a role in reducing the risk of ovarian cancer.